

1. Design of an Automatic Traffic Light Controller using Verilog HDL

You are tasked with designing and implementing a robust automatic traffic light controller using Verilog HDL. The system will be deployed at a busy T-junction in Guwahati main road to manage traffic between a main road and a side road. Each road is controlled by a standard set of **Red**, **Yellow**, and **Green** lights. Your design must ensure **safe and efficient** vehicle movement during peak hours.

Road Configuration

- **Main Road (R1):** High priority, experiencing heavy traffic.
- **Side Road (R2):** Lower priority, with lighter traffic.

Timing Sequence

The operational cycle is governed by the following time intervals for the green and yellow phases:

- Main Road Green: **30 seconds**
- Main Road Yellow: **5 seconds**
- Side Road Green: **20 seconds**
- Side Road Yellow: **5 seconds**

It is the designer's responsibility to develop a complete and safe state machine that correctly sequences the lights for both roads based on these durations. The implementation must inherently prevent any hazardous signal combinations.

Inputs

- **clk:** The main system clock for all timing operations.
 - **reset:** A signal to initialize the controller to a default, safe starting state.
 - **sensor:** An optional input from a vehicle detector on the Side Road (R2). A smart implementation should use this signal to skip the Side Road's green phase if no vehicles are waiting, thereby improving traffic flow on the Main Road.
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2. Timing Counter Design using Verilog HDL

Design a counter module that generates the precise time delays needed for the traffic light controller.

Specifications

- **Input Clock:** 50 Hz
- **Required Durations:** The counter must be capable of timing intervals of **5 seconds**, **20 seconds**, and **30 seconds**.

Your module should use the 50 Hz clock to signal the main controller when each specified time duration has completed.

3. Design the top-level Verilog module that integrates your FSM and the timing counter. Show how you would instantiate both sub-modules and connect their ports to create the complete, functional traffic light controller.

Analyze the post-implementation area (LUTs & FFs), power, maximum operating frequency, and timing (delay) report for each of the three design modules on the selected FPGA.